TRƯỜNG ĐẠI HỌC FPT

Requirement & Design Specification

Global Assess Management System (GAMS)

Version: 1.0

– Hanoi, November 2025 –

Record of Changes

Version	Date	A* M, D	In charge	Change Description
V1.0	15/2	A	KienNTHE11	

*A - Added M - Modified D - Deleted

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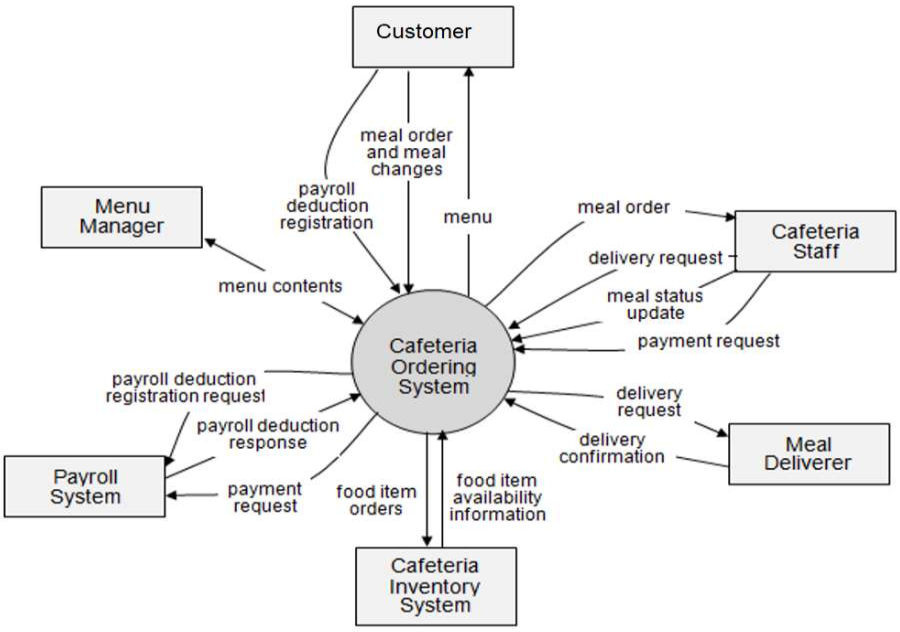
I. Overview

1. System Context

[Gives the overall description about the product with some introduction and the context diagram. The context diagram presents the boundary and connections between the system you're developing and everything else in the universe. This identifies external entities (or terminators – software, hardware, human components, and other systems) outside the system that interface to it in some way, as well as data, control, and material flows between the terminators and the system.]

<<Sample: The Cafeteria Ordering System is a new software system that replaces the current manual and telephone processes for ordering and picking up meals in the Process Impact cafeteria. The context diagram below illustrates the external entities and system interfaces for release 1.0. The system is expected to evolve over several releases, ultimately connecting to the Internet ordering services for several local restaurants and to credit and debit card authorization services.



>>

2. External Entities

[Describe the external entities who/what might interact with your system as mentioned in the context diagram above, using the form of table as following]

#	Entity	Description
1	Customer	Entity description here..
2	Cafeteria Staff	..
3	…	

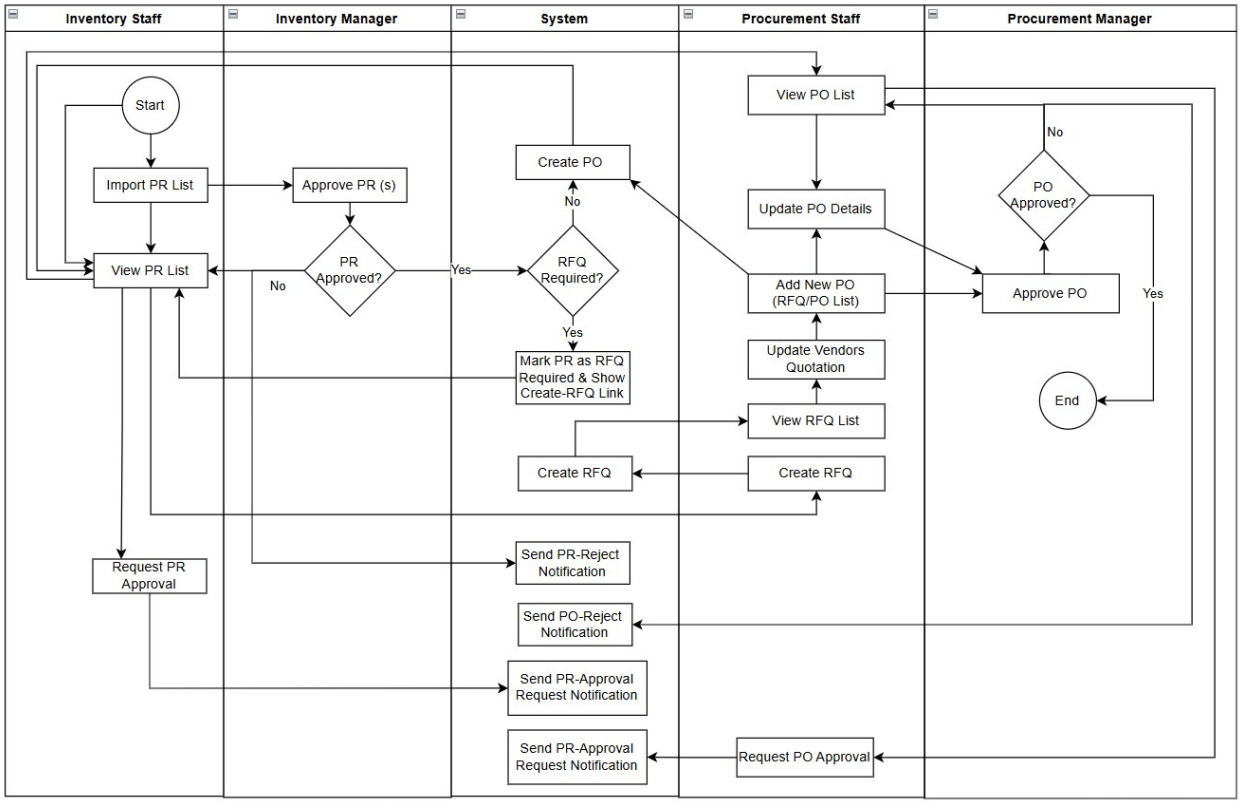
3. Business Processes

[Provide business processes () using the swim-lane diagram(s) like sample below

A business process a collection of activities that takes one or more kinds of input and creates an output that is of value to the customer. The customer for a business process may be a traditional external

customer (the person who buys the finished product), or it may be an internal customer (such as a colleague in another department)]

3.1 Sale Process



3.2 Customer Support

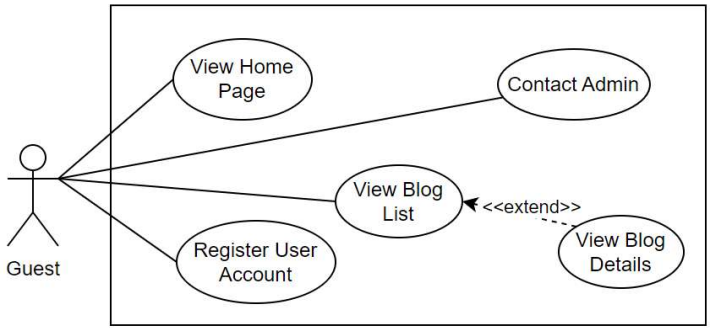
…

4. User Requirements

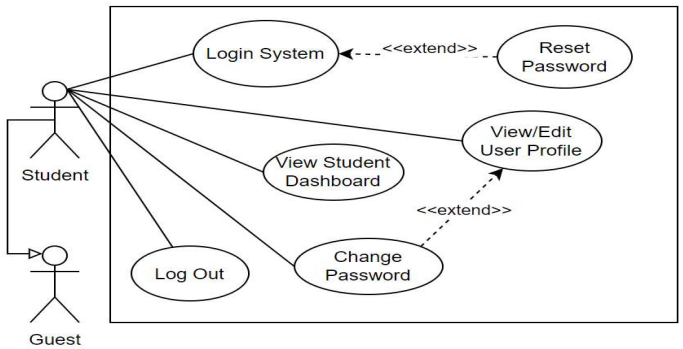
In this section, you need to provide the descriptions for all the use case (UC) you would have identified (in the form of table), and UC diagram(s) to show the actor-UCs and UC-UC relationships (like the sample below] – just one consolidated diagram or multiple diagrams for the system, each diagram is for one actor or one workflow]

ID	Use Case	Feature	Main Actor	Use Case Description
01	View Menu	Menu Browsing	Customer	Customer can browse the available meals, drinks, and combos with details like description, price, and availability
02	…			

4.1 UCs for Guest



4.2 UCs for Student



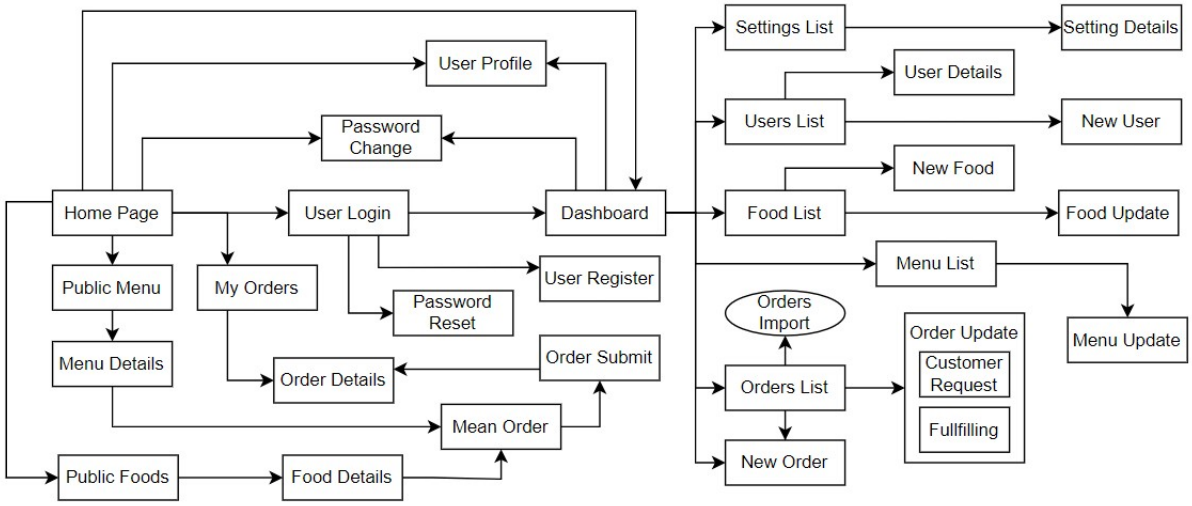
4.3 …

5. System Functionalities

[Provide functionality overview of software system: screen flow, screen descriptions, system user roles, screen authorization, non-screen functions, ERD]

5.1 Screens Flow

[This part shows the system screens and the relationship among screens. You can draw the Screens Flow for the system in the form of diagram as below.]



5.2 Screen Authorization

[Provide the system roles authorization to the system features (down to screens, and event to the screen activities if applicable) in the table form as below – replace Role-Name1, Role-Name2,… with your specific system user role names]

Screen	Role-Name1	Role-Name2	Role-Name3	…
<<Screen Name1>>	X		X	X
<<Screen Activity>>			X	X
<<Screen Name2>>	X		X	
Query All Data	X			
Query Own Data			X	
Add New Data			X	X
Update All Data				X
Update Own Data				X
Delete Data				
…				

5.3 Non-UI Functions

[Provide the descriptions for the non-screen system functions, i.e batch/cron job, service, API, etc.]

#	Feature	System Function	Description
1	<<Feature Name>>	<<Function Name1>>	<<Function Name1 Description>>
2	…		

II. Functional Requirements

[Provide descriptions about the system's functions/screens. The functions/screens are grouped by the system features, and even sub-features if needed. For the screens, you need to provide the screen layouts (mock-up screens) and relevant specifications if needed]

1. Feature Name1

1.1 SubFeature Name1.1

1.1.1 Screen/Function Name1

[Content #1: UI layout (Mockup screen prototype)]

[Content #2: brief descriptions of the screen/function, mapped to the relevant use cases]

[Content #3: provide further descriptions for the screen's components/fields using table format below]

Field Name	Description
Field Name1	Field description: data type min/max length or value, initial data, etc.
Field Name2	…
Field Group-Name1	
Field Name3	…
Field Name4	…
Field Group-Name2	
…	…

1.1.2 Screen/Function Name2

…

1.2 SubFeature Name1.2

…

2. User Authentication

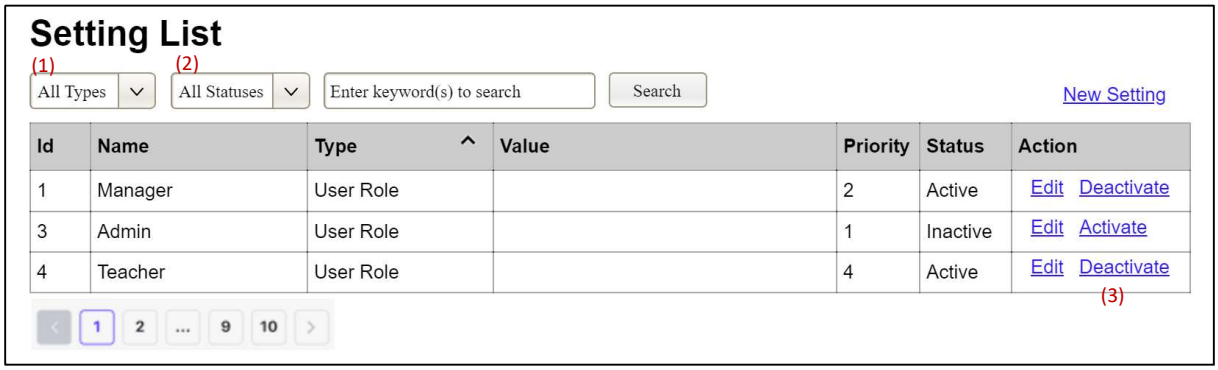
2.1 User Register

…

3. System Administration

3.1 Master Data

3.1.1 Setting List



This screen allows the Administrator to:

- View Setting List: view list of current master data.

- Filter Setting List: filter master data by data types, statuses
- Search Settings: enter keyword(s) to search master data by their names or values
- Sort Setting List: sort mater data list (ascending, descending) by clicking column headers

On the screen, s/he can also

- Activate/Deactivate Setting: change status of a specific inactive/active master data
- Choose to go to the Setting Details screens for adding new or updating an existing master data by clicking the New Setting or Edit link.

Field Description

Field Name	Description
(1)	Initial values: all the active setting names with null or blank type Hover the mouse to show the field name: "Setting Type"
(2)	Initial values: All Statuses, Active, Inactive (default value "All Status") Hover the mouse to show the field name: "Setting Status"
(3)	The change-status action is Activate or Deactivate depending on the current status of the relevant setting (Inactive or Active, respectively).

3.1.2 Setting Details

This screen allows the Administrator to:

- Add New Setting: add new master data.
- Update Setting Details: update details of a specific master data

Field Description

Field Name	Description
Name	Data type: non-digit string, max length of 20 characters
Type	Initial data values: all active setting names (with null or blank type)
Value	Data type: any string, max length of 100 characters
Priority	Data type: a positive integer
Description	Data type: any string, max length of 200 characters

3.2 User Management

3.2.1 User List

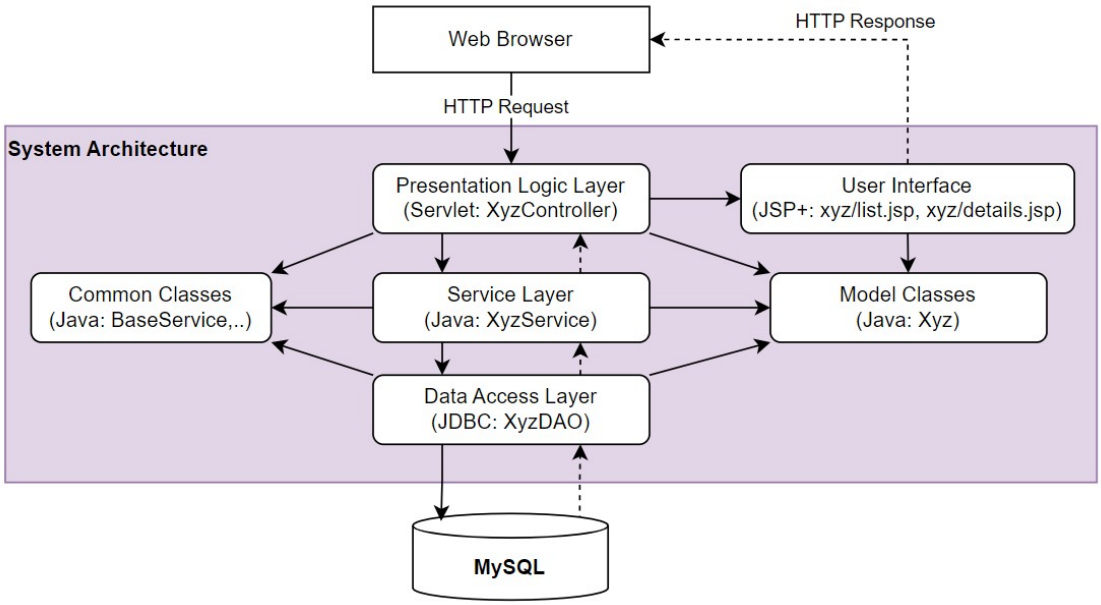
…

III. System Design

1. Software Architecture

[The content of this section includes the overall architectural diagram which includes the sub-systems and/or components, the external systems (if any), and the relationships (communication messages) among them. You need also provide the explanation for each of the diagram components (modules, sub-systems, external systems, etc.)].

1.1 Overall Architecturural Diagram



1.2 Component Descriptions

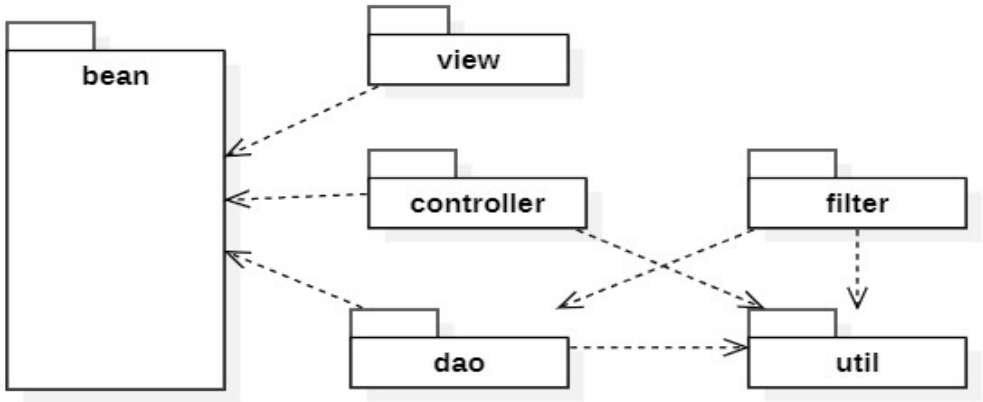
No	Component	Description
01	*Presentation Logic Layer*	*<Description of the component>*
02	*User Interface*	*<Description of the package>*
03	*…*	

-

2. Code Package Design

[Provide the package diagram for each sub-system. The content of this section including the overall package diagram, the explanation, package and class naming conventions in each package. Please see the sample & description table format below]

2.1 Package Diagram



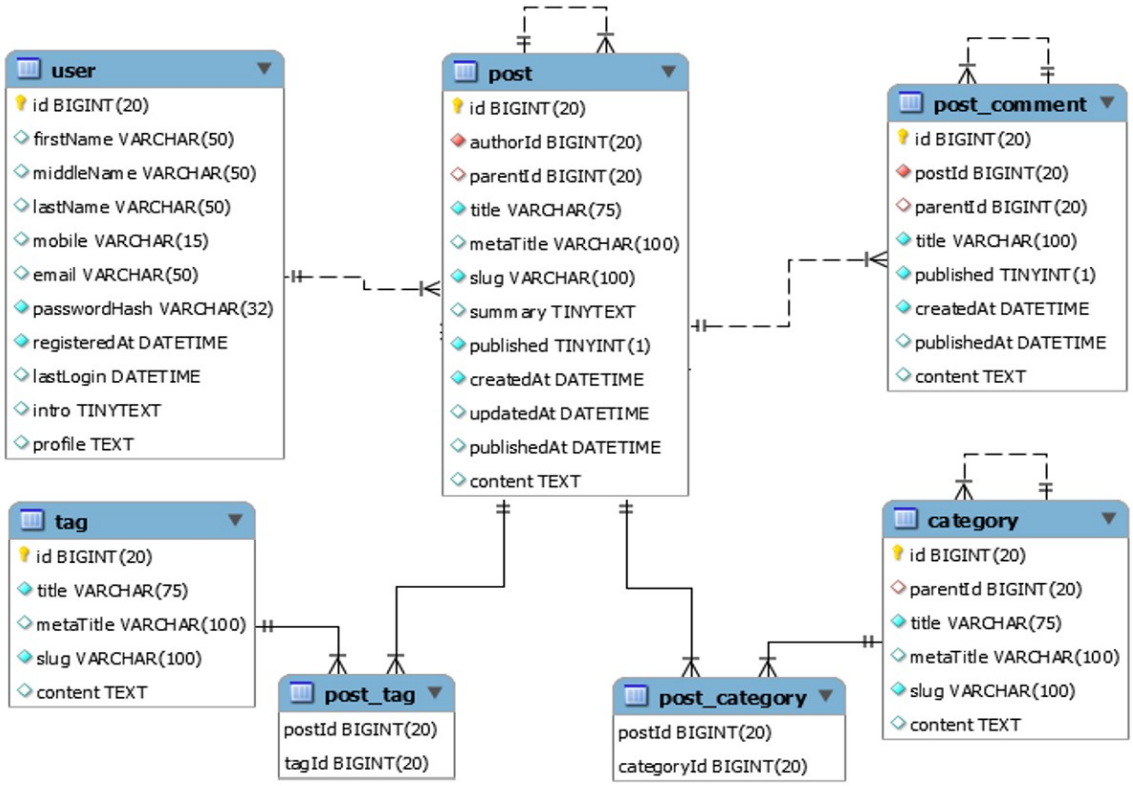
2.2 Package Descriptions

No	Package	Description
01	*View*	*<Description of the package>*
02	*bean*	*<Description of the package>*
03	*…*	

3. Database Design

[Provide the files description, database table relationship & table descriptions like example below]

3.1 Database Schema



3.2 Table Descriptions

3.2.1 table_name1

[Provide brief description of the table here]

[Provide the detailed table fields description using below table format

* PK~Primary Key; FK~Foreign Key; UN~Unique; NN ~ not null

No	Field	PK	FK	UN	NN	Description
01	*<field_name1>*					*<Description of the field_name1>*
02	*<field_name2>*					…
..						

3.2.2 table_name2

…